

# SANKALP SHANKAR

📞 9153946929 ✉ sankalpshankar02@gmail.com

🌐 [linkedin.com/in/sankalp-shankar-734007284/](https://www.linkedin.com/in/sankalp-shankar-734007284/)

🐙 [github.com/sankalp-happy](https://github.com/sankalp-happy)

🔗 [leetcode.com/sankalpshankar/](https://leetcode.com/sankalpshankar/)

## Objective

ML Engineer who builds and ships LLM systems end-to-end — RAG pipelines, model serving, and the FastAPI/React apps around them. I like owning a project from idea to production.

## Education

**Acharya Institute of Technology**

*Bachelor of Engineering in AIML*

- SGPA: 8.7/10.0

**Expected July 2027**

*Bangalore, India*

## Experience

**Kapture CX**

**June 2026 – Present**

*Machine Learning Intern*

*Bangalore, India*

- Tuned vLLM serving flags and restructured prompts for prefix-cache reuse, raising concurrency  $\sim 15 \rightarrow \sim 40$  RPS (benchmark) and  $\sim 12-15 \rightarrow \sim 25$  RPS in production
- Benchmarked GPTQ/AWQ quantization and vLLM configs for Gemma 4 31B-IT on H200 GPUs
- Recommended vLLM-router over nginx round-robin for cache-aware routing; deployed vLLM as a systemd service in production

**RDSO Documents – Southern Railway (Freelance)**

**March – June 2026**

*ML/Backend Engineer*

*Remote*

- Built an ETag-based crawler to detect revised engineering standards, since the source portal reuses URLs on revision
- Synced only changed rows to the outside network on a weekly cron; archived superseded documents instead of deleting them

## Technical Skills

**Languages & Frameworks:** Python, JavaScript, HTML5, CSS, React, FastAPI, Express.js, Node.js

**AI / LLM:** OpenAI, Gemini, LangChain, RAG, Vector Databases, vLLM, TensorFlow, PyTorch, XGBoost, Scikit-learn, NLTK

**Data & Databases:** NumPy, Pandas, SQL, MongoDB, Redis

**Cloud & DevOps:** AWS, Docker, CI/CD, Git, GitHub, Jupyter Notebook, VS Code

## Projects

**PharmaGuard** | *LLM-Enabled Clinical Decision Support System* | [live link](#)

**2026**

- Built a FastAPI + React platform that turns genomic VCF files into CPIC-guided prescribing recommendations via a deterministic, PharmCAT-integrated risk engine
- Kept Llama 3.3 (via Groq) confined to explaining a recommendation already made, so a model failure never changes the decision
- Exported EHR-ready JSON; containerized and deployed on AWS with Docker

**SwitchBoard** | *LLM Gateway & Multi-Provider Routing Layer* | [live link](#)

**2026**

- Built an OpenAI-compatible gateway routing across Groq, Google, and Anthropic with automatic failover and quota-aware key rotation
- Added a Redis semantic cache (Gemini embeddings, cosine similarity) that serves rephrased repeat questions with zero provider calls
- Made provider switching a config change, not a code change; monitored with Prometheus/Grafana

**MediAssist AI** | *RAG-Based Agentic CDSS — Hackathon Runner-up* | [GitHub repo](#)

**2025**

- Built a RAG pipeline using LangChain and a vector database for structured medical knowledge retrieval with OpenAI-backed agents
- Integrated XGBoost predictive models for explainable differential diagnosis with confidence-scored outputs
- Deployed full-stack application with an automated NLP feedback loop for continuous output improvement

## Achievements & Leadership

**Technical Achievements**

**2022 – Present**

*Competitive Programming, Hackathons & National Tech Events*

*Multiple Platforms*

- 7× Competition Winner: 4 Hackathon wins (1st – State Level, Runner-up – 48hr, 3rd – Zonal) and 2 National AI Ideathon titles;  $\sim 1600$  LeetCode rating, 260+ problems solved

**Vice President, ‘The Big O’ Tech Club & Event Head, Tech Habba**

**2023 – Present**

*Acharya Institute of Technology*

*Bangalore, India*

- Led Tech Habba 2024, a 3-day institute-wide festival, and a 2-day, 500+ participant techfest; mentored junior developers along the way

## Certifications

- Deep Learning (NPTEL – IIT Ropar): Secured 84% in 12-week course
- Meta Professional Certificate in Front-End Development - 9-course curriculum with capstone project